

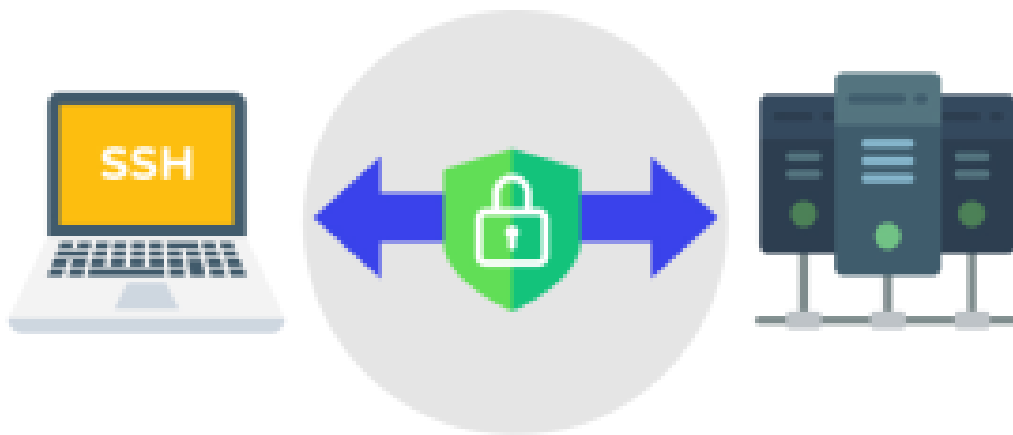
Administración Remota de Servidores Web en AWS mediante SSH

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Arquitectura en la nube

2º ASIR

Protocolo de shell seguro



Índice

Administración Remota de Servidores Web en AWS mediante SSH.....	1
Índice.....	2
1. Introducción y Conceptos Básicos.....	4
¿Qué es AWS?.....	4
¿Qué es SSH?.....	4
Archivos PEM: Tu Llave de Acceso.....	4
Security Groups: El Firewall de AWS.....	4
2. Preparación del Entorno Local.....	5
Paso 1: Verificar WSL2.....	5
Paso 2: Crear Directorio para Claves SSH.....	6
Paso 3: Verificar Conectividad a Internet.....	6
3. Configuración en AWS.....	6
Fase 1: Obtener la Clave PEM.....	6
Paso 1.1: Descargar desde el Laboratorio AWS.....	6
Paso 1.2: Mover la Clave a WSL.....	6
Paso 1.3: Configurar Permisos de Seguridad.....	7
Fase 2: Crear Instancia EC2.....	7
Paso 2.1: Acceder a EC2.....	8
Paso 2.2: Configuración Básica.....	8
Paso 2.3: Configuración de Red.....	10
Fase 3: Configurar Security Group.....	11
Paso 3.1: Crear o Editar Security Group.....	11
Paso 3.2: Configurar Reglas de Entrada.....	11
Paso 3.3: Lanzar la Instancia.....	13
Fase 4: Obtener la IP Pública.....	13
4. Conexión SSH.....	13
Paso 1: Verificar Conectividad Básica.....	13
Paso 2: Conectar por SSH.....	13
Paso 3: Aceptar la Huella Digital.....	14
Paso 4: Verificar Conexión Exitosa.....	14
5. Verificación y Pruebas.....	14
Verificar Información del Sistema.....	14
Actualizar el Sistema.....	16
Instalar Herramientas Básicas.....	17
Verificar Security Group desde el Servidor.....	17
6. Solución de Problemas.....	18
Error: "Permission denied (publickey)".....	18
Error: "Connection refused".....	18
Error: "Connection timed out".....	19
Error: "Host key verification failed".....	19
Usuario Incorrecto.....	19
Comandos de Referencia Rápida.....	19

Gestión de Claves PEM.....	19
Conexión SSH.....	20
Verificación de Conectividad.....	20
Próximos Pasos.....	20
Buenas Prácticas de Seguridad.....	21
Para la Clave PEM.....	21
Para Security Groups.....	21
Para las Instancias.....	21
Recursos Adicionales.....	21
Documentación Oficial.....	21
Comandos Útiles de Linux.....	22
Glosario.....	22
Repetición de práctica anterior (CONFIGURACIÓN AVANZADA DE SERVIDORES WEB Y HTTPS).....	23

1. Introducción y Conceptos Básicos

¿Qué es AWS?

AWS (Amazon Web Services) es una plataforma de servicios en la nube que permite alquilar recursos computacionales sin necesidad de infraestructura física propia. En lugar de comprar y mantener servidores, AWS proporciona instancias virtuales (EC2) que funcionan como ordenadores completos en la nube.

Ventajas principales:

- Pago por uso (solo pagas lo que consumes)
- Escalabilidad instantánea
- Alta disponibilidad
- Mantenimiento gestionado por Amazon

¿Qué es SSH?

SSH (Secure Shell) es un protocolo de red que permite acceder de forma segura a servidores remotos. Toda la comunicación viaja cifrada, protegiendo tus datos de interceptaciones.

Características clave:

- Cifrado de extremo a extremo
- Autenticación mediante claves criptográficas
- Ejecución remota de comandos
- Transferencia segura de archivos

Archivos PEM: Tu Llave de Acceso

Un archivo `.pem` contiene una clave privada que funciona como tu identificación única para acceder al servidor.

Conceptos importantes:

- **Clave privada (.pem):** Solo tú la tienes. NUNCA la compartas.
- **Clave pública:** AWS la guarda en el servidor.
- **Principio de funcionamiento:** Autenticación mediante criptografía asimétrica.

Reglas de seguridad:

- Guarda el archivo en ubicación segura
- Establece permisos restrictivos (400)
- Nunca subas la clave a repositorios públicos
- Crea copias de seguridad cifradas

Security Groups: El Firewall de AWS

Los Security Groups controlan el tráfico de red hacia y desde tu instancia EC2, actuando como un firewall virtual.

Puertos que configuraremos:

- **Puerto 22 (SSH):** Acceso administrativo remoto
- **Puerto 8080:** Servidor web Apache
- **Puerto 8081:** Servidor web Nginx
- **Puerto 8082:** Servidor web Caddy
- **Puerto 8443:** Apache con HTTPS (SSL/TLS)

2. Preparación del Entorno Local

Paso 1: Verificar WSL2

Abre PowerShell o CMD en Windows y ejecuta:

```
wsl --version
```

Resultado esperado:

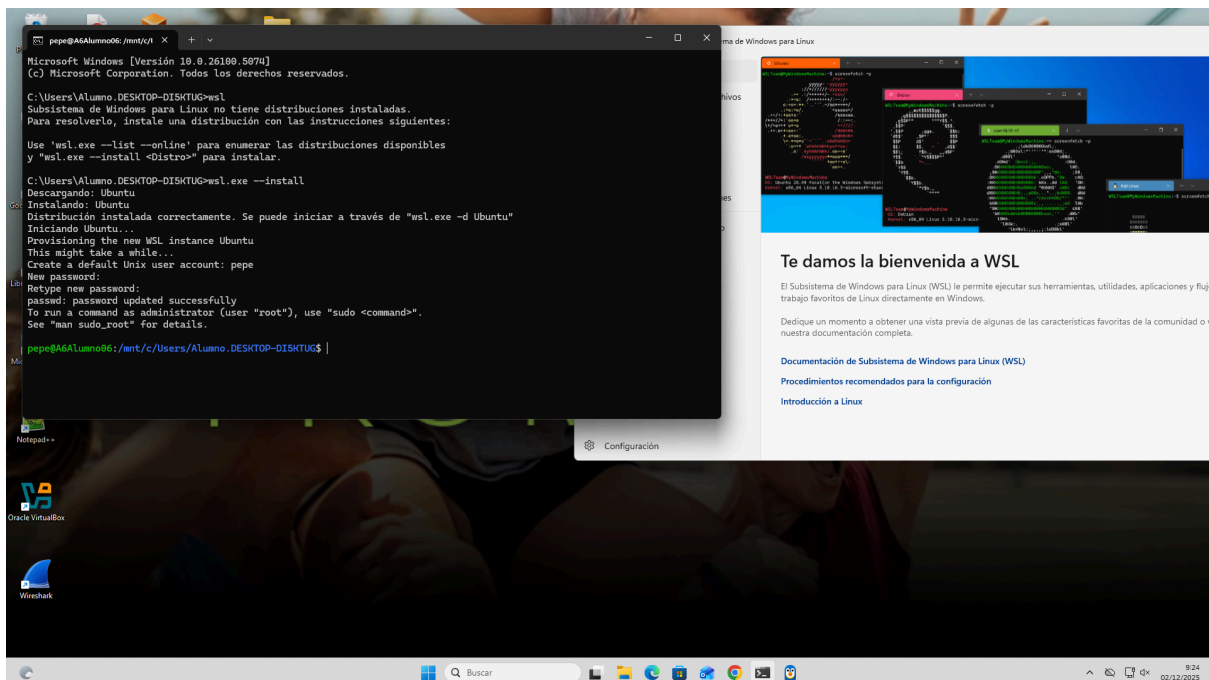
Versión de WSL: 2.x.x.x

Versión de kernel: 5.x.x.x

Versión de WSLg: 1.x.x

Si WSL no está instalado, ejecuta:

```
wsl --install
```



Paso 2: Crear Directorio para Claves SSH

Abre tu terminal WSL (Ubuntu) y ejecuta:

```
# Crear directorio .ssh si no existe
mkdir -p ~/.ssh

# Establecer permisos correctos (solo el propietario puede acceder)
chmod 700 ~/.ssh

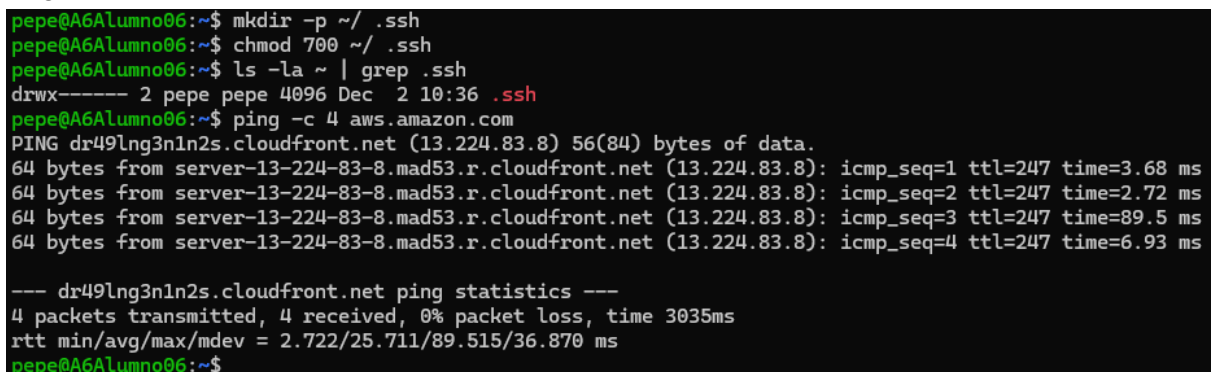
# Verificar que se creó correctamente
ls -la ~ | grep .ssh
```

Resultado esperado:

```
drwx----- 2 usuario usuario 4096 dic  2 10:30 .ssh
```

Paso 3: Verificar Conectividad a Internet

ping -c 4 aws.amazon.com



```
pepe@A6Alumno06:~$ mkdir -p ~/.ssh
pepe@A6Alumno06:~$ chmod 700 ~/.ssh
pepe@A6Alumno06:~$ ls -la ~ | grep .ssh
drwx----- 2 pepe pepe 4096 Dec  2 10:36 .ssh
pepe@A6Alumno06:~$ ping -c 4 aws.amazon.com
PING dr49lng3n1n2s.cloudfront.net (13.224.83.8) 56(84) bytes of data.
64 bytes from server-13-224-83-8.mad53.r.cloudfront.net (13.224.83.8): icmp_seq=1 ttl=247 time=3.68 ms
64 bytes from server-13-224-83-8.mad53.r.cloudfront.net (13.224.83.8): icmp_seq=2 ttl=247 time=2.72 ms
64 bytes from server-13-224-83-8.mad53.r.cloudfront.net (13.224.83.8): icmp_seq=3 ttl=247 time=89.5 ms
64 bytes from server-13-224-83-8.mad53.r.cloudfront.net (13.224.83.8): icmp_seq=4 ttl=247 time=6.93 ms

--- dr49lng3n1n2s.cloudfront.net ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 2.722/25.711/89.515/36.870 ms
pepe@A6Alumno06:~$
```

3. Configuración en AWS

Fase 1: Obtener la Clave PEM

Paso 1.1: Descargar desde el Laboratorio AWS

1. Accede a tu consola del laboratorio AWS
2. Localiza el botón **"Download PEM"** o **"Download key"**
3. Descarga el archivo `labsuser.pem`
4. Anota la ubicación donde se guardó

Paso 1.2: Mover la Clave a WSL

Opción A: Desde la carpeta de Descargas de Windows

Reemplaza TU-USUARIO con tu nombre de usuario de Windows

```
cp /mnt/c/Users/TU-USUARIO/Downloads/labsuser.pem ~/.ssh/
```

```
pepe@A6Alumno06: /mnt/c/Users/Alumno.DESKTOP-DI5KTUG$ cp /mnt/c/Users/Alumno.DESKTOP-DI5KTUG/Downloads/labsuser_
_pepe.pem ~/.ssh/
```

Opción B: Si ya está en WSL

```
cp ~/Downloads/labsuser.pem ~/.ssh/
```

Paso 1.3: Configurar Permisos de Seguridad

Establecer permisos restrictivos (solo lectura para el propietario)

```
chmod 400 ~/.ssh/labsuser.pem
```

Verificar permisos

```
ls -la ~/.ssh/labsuser.pem
```

Resultado esperado:

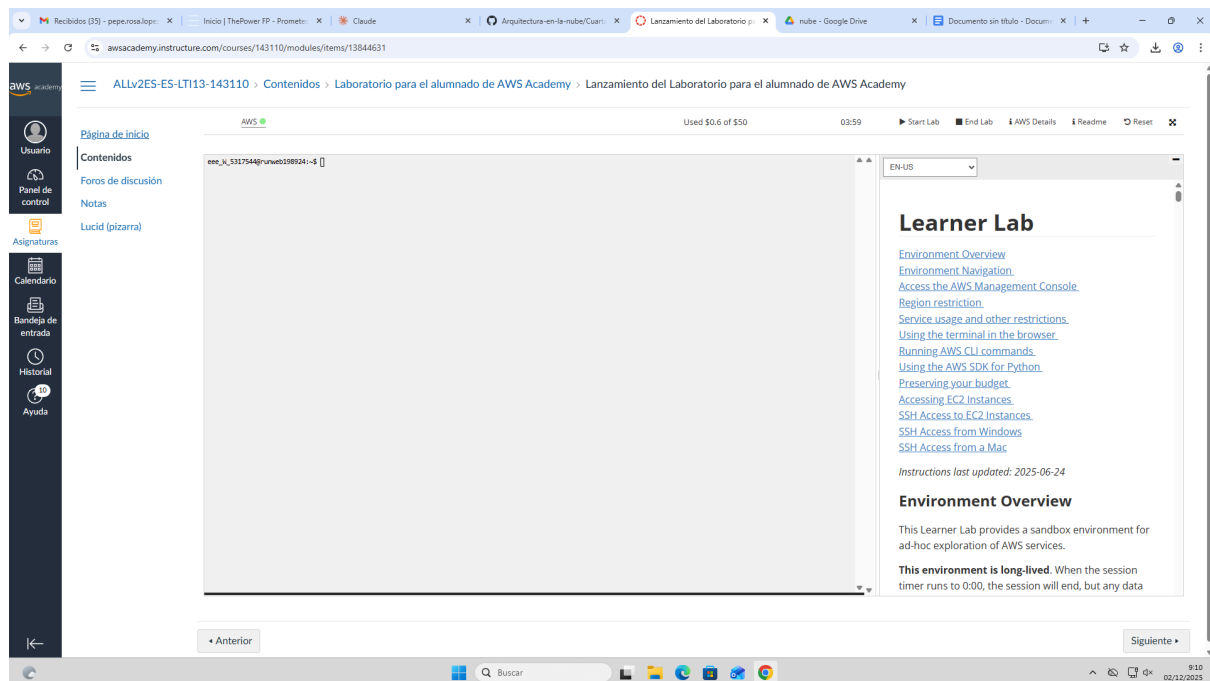
```
-r----- 1 usuario usuario 1674 dic  2 10:35 /home/usuario/.ssh/labsuser.pem
```

```
pepe@A6Alumno06: /mnt/c/Users/Alumno.DESKTOP-DI5KTUG$ chmod 400 ~/.ssh/labsuser_pepe.pem
pepe@A6Alumno06: /mnt/c/Users/Alumno.DESKTOP-DI5KTUG$ ls -la ~/.ssh/labsuser_pepe.pem
-r----- 1 pepe pepe 1674 Dec  2 10:17 /home/pepe/.ssh/labsuser_pepe.pem
pepe@A6Alumno06: /mnt/c/Users/Alumno.DESKTOP-DI5KTUG$
```

IMPORTANTE: Si los permisos son incorrectos (por ejemplo, `-rw-r--r--`), SSH rechazará la conexión por razones de seguridad.

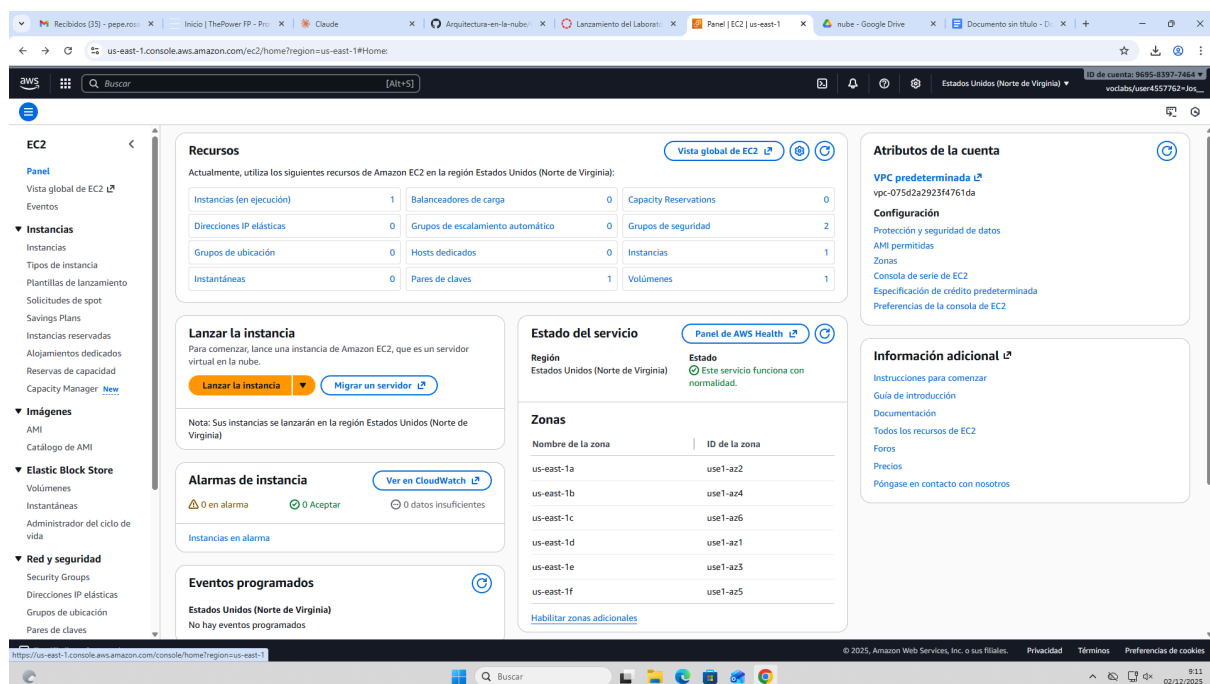
Fase 2: Crear Instancia EC2

The screenshot shows the AWS Academy Learner Lab interface. The browser address bar displays the URL: `awsacademy.instructure.com/courses/143110/modules/items/13844631`. The page title is "Lanzamiento del Laboratorio para el alumnado de AWS Academy". The interface includes a sidebar with navigation options: "Página de inicio", "Contenidos", "Foros de discusión", "Notas", "Lucid (pizarra)", "Asignaturas", "Calendario", "Bandeja de entrada", "Historial", and "Ayuda". The main content area shows a terminal window with the command `aws ec2 describe-instances --profile awsacademy --region us-east-1 --query 'Instances[*].InstanceId'` and its output: `["i-03375440runet398924"]`. The right sidebar contains the "Learner Lab" section with links to "Environment Overview", "Environment Navigation", "Access the AWS Management Console", "Region restriction", "Service usage and other restrictions", "Using the terminal in the browser", "Running AWS CLI commands", "Using the AWS SDK for Python", "Preserving your budget", "Accessing EC2 Instances", "SSH Access to EC2 Instances", "SSH Access from Windows", and "SSH Access from a Mac". The "Environment Overview" section states: "This Learner Lab provides a sandbox environment for ad-hoc exploration of AWS services. This environment is long-lived. When the session timer runs to 0:00, the session will end, but any data".



Paso 2.1: Acceder a EC2

1. En la consola AWS, busca **"EC2"** en la barra de búsqueda
2. Haz clic en **"Launch Instance"** (Lanzar instancia)



Paso 2.2: Configuración Básica

Nombre y etiquetas:

Nombre: servidor-web-practica

Etiqueta adicional (opcional):

Key: Curso

Value: ArquitecturaNube

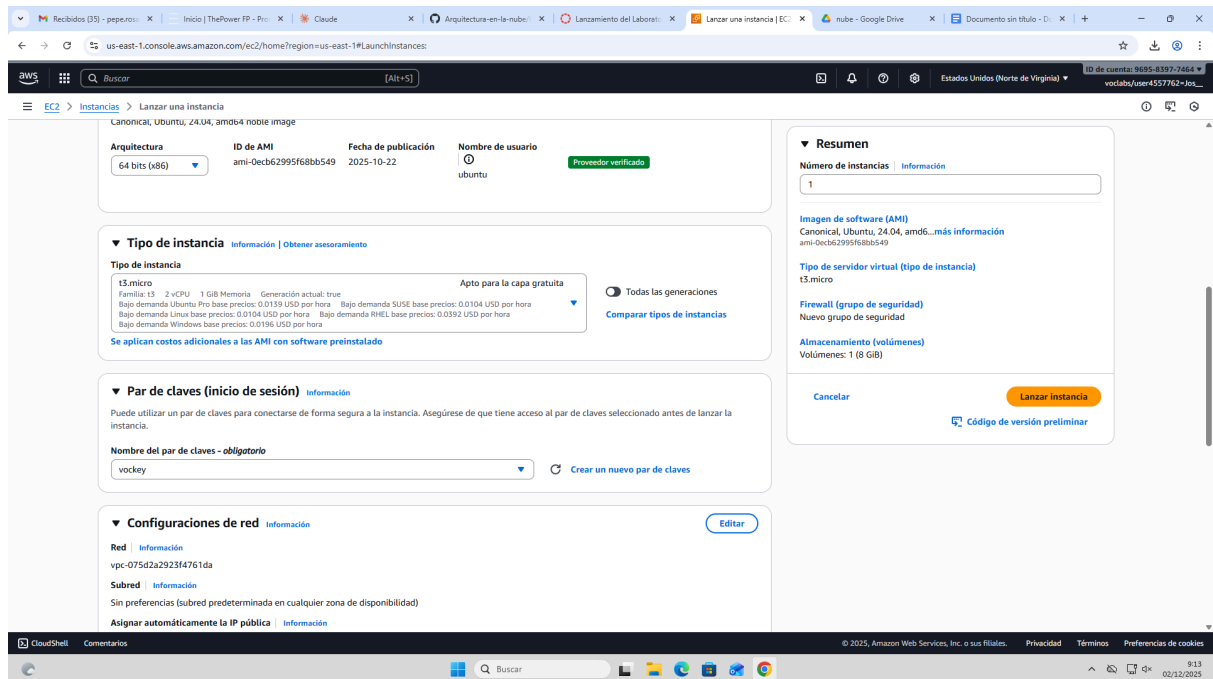
AMI (Amazon Machine Image):

- Selecciona: **Ubuntu Server 22.04 LTS** o **Ubuntu 24.04 LTS**
- Arquitectura: **64-bit (x86)**

Tipo de instancia:

- Selecciona: **t2.micro** o **t3.micro** (Free Tier eligible)
- vCPUs: 1
- Memoria: 1 GiB

The screenshot shows the AWS Management Console 'Lanzar una instancia' (Launch Instance) page. The browser tabs include 'Recibidos (35) - pepenro...', 'Inicio | ThePowerTP - P...', 'Claude', 'Arquitectura-en-la-nube', 'Lanzamiento del Labor...', 'Lanzar una instancia | EC...', 'nube - Google Drive', and 'Documento sin título - D...'. The AWS console header shows the 'us-east-1' region and the user's account ID. The main content area is titled 'Lanzar una instancia' and includes a 'Nombre y etiquetas' section with a text input for the instance name. Below this is the 'Imágenes de aplicaciones y sistemas operativos' section, which features a search bar and a grid of AMI options including Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. The 'Ubuntu' option is highlighted. To the right, a 'Resumen' (Summary) panel displays the selected configuration: 1 instance, Canonical Ubuntu 24.04 AMI, t3.micro instance type, new security group, and 1 GB storage. At the bottom right of the summary panel are 'Cancelar' and 'Lanzar instancia' buttons. The footer of the console shows the copyright notice for Amazon Web Services, Inc. and the date 02/12/2023.



Paso 2.3: Configuración de Red

Par de claves:

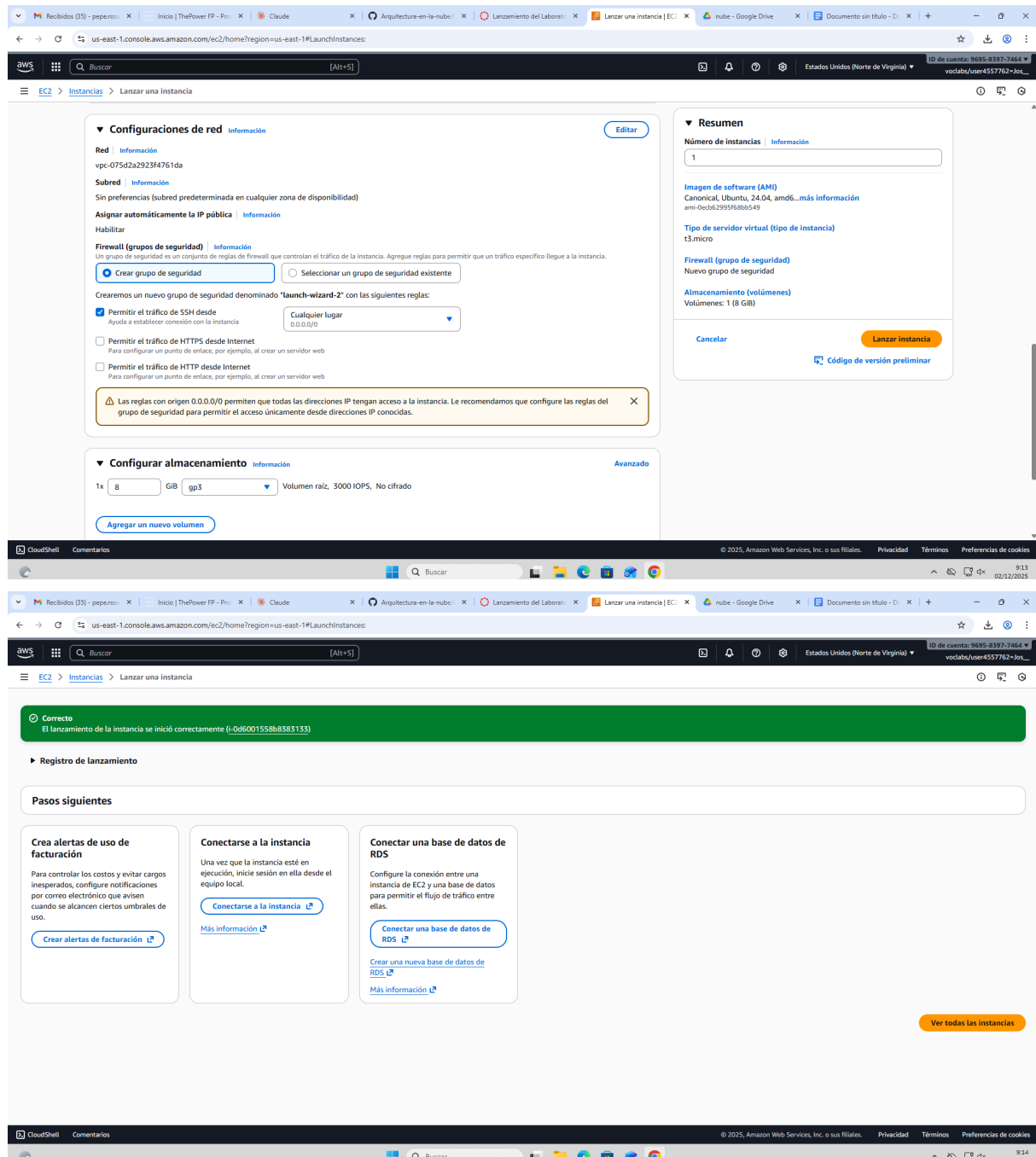
- Selecciona: **"Choose an existing key pair"**
- Nombre: **labsuser**
- ⚠ Si no aparece, asegúrate de haber descargado correctamente el archivo PEM

Configuración de red:

- VPC: **default**
- Subnet: **No preference** (default)
- Auto-assign public IP: **Enable** ✓

Almacenamiento:

- Tamaño: **8 GiB**
- Tipo: **gp2** (General Purpose SSD)



Fase 3: Configurar Security Group

Paso 3.1: Crear o Editar Security Group

Nombre: **sg-servidores-web**

Descripción: Security group para servidores web con Apache, Nginx y Caddy

Paso 3.2: Configurar Reglas de Entrada

Añade las siguientes 5 reglas:

Tipo	Protocolo	Puerto	Origen	Descripción
SSH	TCP	22	0.0.0.0/0	Acceso SSH administrativo
Custom TCP	TCP	8080	0.0.0.0/0	Apache HTTP
Custom TCP	TCP	8081	0.0.0.0/0	Nginx
Custom TCP	TCP	8082	0.0.0.0/0	Caddy
Custom TCP	TCP	8443	0.0.0.0/0	Apache HTTPS

Configuración en la consola AWS:

Inbound rules:

1. Type: SSH | Protocol: TCP | Port: 22 | Source: 0.0.0.0/0
2. Type: Custom TCP | Protocol: TCP | Port: 8080 | Source: 0.0.0.0/0
3. Type: Custom TCP | Protocol: TCP | Port: 8081 | Source: 0.0.0.0/0
4. Type: Custom TCP | Protocol: TCP | Port: 8082 | Source: 0.0.0.0/0
5. Type: Custom TCP | Protocol: TCP | Port: 8443 | Source: 0.0.0.0/0

Nota de seguridad: En producción, **0.0.0.0/0** (acceso desde cualquier IP) debe restringirse a IPs específicas.

Editar reglas de entrada información

Las reglas de entrada controlan el tráfico entrante que puede llegar a la instancia.

ID de la regla del grupo de seguridad	Tipo	Protocolo	Intervalo de puertos	Origen	Descripción: opcional	
sgr-06791e9e7edf55ed3	SSH	TCP	22	Anywh...		Eliminar
-	TCP personalizado	TCP	8080	Anywh...	0.0.0.0/0	Eliminar
-	TCP personalizado	TCP	8081	Anywh...	0.0.0.0/0	Eliminar
-	TCP personalizado	TCP	8082	Anywh...	0.0.0.0/0	Eliminar
-	TCP personalizado	TCP	8084	Anywh...	0.0.0.0/0	Eliminar

[Agregar regla](#)

⚠ Las reglas cuyo origen es 0.0.0.0/0 o ::/0 permiten a todas las direcciones IP acceder a la instancia. Recomendamos configurar reglas de grupo de seguridad para permitir el acceso únicamente desde direcciones IP conocidas.

[Cancelar](#) [Previsualizar los cambios](#) [Guardar reglas](#)

Paso 3.3: Lanzar la Instancia

1. Revisa todas las configuraciones
2. Haz clic en "**Launch Instance**"
3. Espera 2-3 minutos hasta que el estado sea "**Running**"

Fase 4: Obtener la IP Pública

1. Ve a **EC2** → **Instances**
2. Selecciona tu instancia `servidor-web-practica`
3. En el panel inferior, localiza "**Public IPv4 address**"
4. Copia esta IP (ejemplo: `54.123.45.67`)

4. Conexión SSH

Paso 1: Verificar Conectividad Básica

Antes de conectar por SSH, verifica que la instancia responde:

```
# Reemplaza TU-IP-PUBLICA con la IP de tu instancia
ping -c 4 TU-IP-PUBLICA
```

Paso 2: Conectar por SSH

```
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

Ejemplo real:

```
ssh -i ~/.ssh/labsuser.pem ubuntu@54.123.45.67
```

```

pepe@A6Alumno06:~$ chmod 400 labsuser.com
chmod: cannot access 'labsuser.com': No such file or directory
pepe@A6Alumno06:~$ cd ~/.ssh
pepe@A6Alumno06:~/.ssh$ chmod 400 labsuser.pem
pepe@A6Alumno06:~/.ssh$ ssh -i ~/.ssh/labsuser.pem ubuntu@3.236.147.88
The authenticity of host '3.236.147.88 (3.236.147.88)' can't be established.
ED25519 key fingerprint is SHA256:Eu7h0kP0n3g7JXHZaPvzn50DuYAaxS89y6H/BN5iFVg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '3.236.147.88' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Dec  9 09:43:01 UTC 2025

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.0% of 6.71GB Processes:    111
Memory usage: 24%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.74.199

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

https://ubuntu.com/aws/pro

```

Paso 3: Aceptar la Huella Digital

La primera vez verás un mensaje como:

The authenticity of host '54.123.45.67 (54.123.45.67)' can't be established.
 ECDSA key fingerprint is SHA256:xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx.
 Are you sure you want to continue connecting (yes/no/[fingerprint])?

Escribe **yes** y presiona Enter.

Paso 4: Verificar Conexión Exitosa

Deberías ver algo como:

Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-1045-aws x86_64)

```

 * Documentation: https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

```

ubuntu@ip-172-31-xx-xx:~\$

5. Verificación y Pruebas

Verificar Información del Sistema

Información del sistema operativo

lsb_release -a

Información del kernel

uname -a

Información de recursos

free -h

df -h

Dirección IP privada

hostname -I

Usuario actual

whoami

```
ubuntu@ip-172-31-74-199: ~  
ubuntu@ip-172-31-74-199:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.3 LTS  
Release:        24.04  
Codename:       noble  
ubuntu@ip-172-31-74-199:~$ uname -a  
Linux ip-172-31-74-199 6.14.0-1015-aws #15~24.04.1-Ubuntu SMP Tue Sep 23 22:44:48 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
ubuntu@ip-172-31-74-199:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:           914Mi       339Mi       342Mi       2.7Mi       389Mi       574Mi  
Swap:           0B           0B           0B  
ubuntu@ip-172-31-74-199:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root        6.8G  1.8G  5.0G  27% /  
tmpfs            458M   0  458M   0% /dev/shm  
tmpfs            183M  876K  182M   1% /run  
tmpfs            5.0M   0   5.0M   0% /run/lock  
efivarfs         128K  3.6K  120K   3% /sys/firmware/efi/efivars  
/dev/nvme0n1p6   881M   89M  730M  11% /boot  
/dev/nvme0n1p5  105M  6.2M   99M   6% /boot/efi  
tmpfs            92M   12K   92M   1% /run/user/1000  
ubuntu@ip-172-31-74-199:~$ hostname -l  
hostname: invalid option -- 'l'  
Usage: hostname [-b] {hostname|-F file}          set host name (from file)  
           hostname [-a|-A|-d|-f|-i|-I|-s|-y]      display formatted name  
           hostname                                display host name  
  
           {yp,nis,}domainname {nisdomain|-F file} set NIS domain name (from file)  
           {yp,nis,}domainname                    display NIS domain name  
  
           dnsdomainname                          display dns domain name  
  
           hostname -V|--version|-h|--help        print info and exit  
  
Program name:  
           {yp,nis,}domainname=hostname -y  
           dnsdomainname=hostname -d  
  
Program options:  
-a, --alias                alias names  
-A, --all-fqdns            all long host names (FQDNs)  
-b, --boot                 set default hostname if none available  
-d, --domain               DNS domain name  
-f, --fqdn, --long         long host name (FQDN)  
-F, --file                 read host name or NIS domain name from given file  
-i, --ip-address           addresses for the host name  
-I, --all-ip-addresses     all addresses for the host  
-s, --short                short host name  
-y, --yp, --nis            NIS/YP domain name  
  
Description:  
This command can get or set the host name or the NIS domain name. You can  
also get the DNS domain or the FQDN (fully qualified domain name).  
Unless you are using bind or NIS for host lookups you can change the  
FQDN (Fully Qualified Domain Name) and the DNS domain name (which is  
part of the FQDN) in the /etc/hosts file.  
ubuntu@ip-172-31-74-199:~$ whoami  
ubuntu  
ubuntu@ip-172-31-74-199:~$
```

Actualizar el Sistema

Actualizar lista de paquetes

sudo apt update

Ver actualizaciones disponibles

apt list --upgradable

Actualizar paquetes

sudo apt upgrade -y

```
ubuntu@ip-172-31-74-199:~$ sudo apt update
```

```
ubuntu@ip-172-31-74-199:~$ sudo apt upgrade -y
```


Instalar Herramientas Básicas

Instalar herramientas de red

```
sudo apt install -y net-tools curl wget
```

Verificar conectividad

```
curl -I https://www.google.com
```

```
ubuntu@ip-172-31-74-199:~$ sudo apt install -y net-tools curl wget
ubuntu@ip-172-31-74-199:~$ curl -I https://www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en"><head><meta content="Search the world's information, including webpages, images, videos and more. Google has many special features to help you find exactly what you're looking for." name="description"><meta content="noodp, " name="robots"><meta content="text/html; charset=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/googleg/1x/googleg_standard_color_128dp.png" itemprop="image"><title>Google</title><script nonce="jpFeSA8dgVosRCP0yo2qAQ">(function(){var _g={kEI:'AfM3adTjAZuy5NoPuaWymQU',kEXPI:'0,1304224,7,2935814,14112,64701,360901,238459,5281866,12368,101,16,16,3,18,36811521,25228681,123988,28395,57126,8040,30632,9140,4599,328,6225,64165,23259,3286,34514,28333,48296,6269,18880,3122,3,2745,7707,7,12148,21237,3927,7,1622,18548,2126,2863,1,5970,7010,9747,5683,3605,20501,7143,6498,2910,8757,1163,7096,3860,1020,1,3316,1,1054,4532,1760,655,5781,9743,2646,98,5,4,1,1164,979,3947,10712,2305,5031,1481,2,1,8108,4914,792,405,185,1,2,1379,136,3355,196,411,3653,1750,2,4205,1,11716,4384,2,873,2,23,2207,5108,3008,26,769,795,3279,4,5074,319,3326,2248,6,6438,1763,2,9016,3322,1031,5337,310,10,525,845,1201,2,11,2,432,1128,3,2842,9,27,8,3055,1009,2,1746,4,4878,5773,3727,2989,14,1,10,4,581,1,1117,389,4,298,2265,84,5,3543,661,5,610,15,164,570,298,2415,1396,499,4943,1275,7,1784,1050,2,295,1619,5,515,665,25,5343,1012,238,4,93,1,421,2139,2,4632,743,4,956,95,646,908,4,794,822,4294,328,605,347,245,4,485,497,249,5,438,4,22,770,1,2,1168,234,3,517,374,360,73,35,2,2141,3,2229,15,516,88,1,8,205,441,795,1,1009,191,4,28,11,3527,16,4,595,2533,1,813,484,5,97,187,201,662,1242,5,179,80,5,197,2366,19,695,751,94,4,47,188,139,1050,127,473,4,16,118,202,164,2199,5,39,218,203,108,6,98,56,5,50,1285,219,4,443,24,975,1389,797,141,398,4,94,888,199,3,2,2,2,341,358,273,153,106,5,3,1106,1254,1,1,91,4,210,395,1,2,2,130,55,267,562,62,95,138,356,2,8,56,99,223,707,191,540,153,10,96,4,32,4,129,144,119,264,3,96,89,14,520,7,616,7,242,371,4,1,1,1,1,74,4,1,39,4,801,871,294,6,534,3,26,17,323,1,378,198,1364,407,1,549,1,177,905,4,97,659,1770,67,3,2,2,2,56,5,958,1011,61,4,173,2,737,2,576,1017,1042,130,4,402,2,203,627,95,1,1,40,651,155,390,30,1,1,1,21,1284,4,141,690,6,1397,2008,610,21011160,9136,1717,7635,5,2992,4,2960,3,2022,6469,2,1558,3,2691,3,10484,1671,24,52,1143,3,739,6,2469,2,4,2,27,518,1000,1469,2,618,626,1,287,112,318,87,275,282,3,87,9,642,516,6492010,119,3001,2005838,974,1625139,915035,10139420,1,450961,4,540768,5,59026,6401,1,112,5,8150,1,1898',kBL:'PxeZ',kOPI:89978449});(function(){var a;((a=window.google)==null?0:a.stvsc)?google.kEI=_g.kEI:window.google=_g;}.call(this));})(function(){google.sn='webhp';google.kHL='en';google.rdn=false;})();(function(){var g=this;self=function k(){return window.google&&window.google.kOPI||null};var l,m=[];function n(a){for(var b;a&&(b=a.getAttribute("eid"))){a=a.parentNode;return b}function q(a){/^http/i.test(a)&&window.location.protocol==='https:'&&(google.ml&&google.ml(Error("a"),!1,{src:a,glmm:1}),a==="");return a}function r(a,b,d,c,h){var e="";b.search("&ei=")===-1&&(e="&ei="+n(c).b.search("&lei=")===-1&&(c=p(c))&&(e+="&lei="+c));var f=b.search("&csid=")===-1&&a!=="slh";c="&z="+Date.now().toString();g._csid&&f&&(c+="&csid="+g._csid);(d=d()&&(c+="&opi="+d);return"/"+(h||"gen_204")+"?atyp=i&ct="+String(a)+"&cad="+b+(e+c));l=google.kEI;google.getEI=n;google.getLEI=p;google.ml=function(){return null};google.log=function(a,b,d,c,h,e){e=e===void 0?k:e.d||((d=r(a,b,e,c,h));i f(d=q(d)){a=new Image;var f=m.length;m[f]=a;a.onerror=a.onload=a.onabort=function(){delete m[f]};a.src=d};google.logUrl=function(a,b){b=b===void 0?k:b;return r("",a,b);}).call(this);(function(){google.y={};google.sy={};function e(a,b,c){if(a)var d=a.id;else{do d=Math.random();while(c[d])}c[d]=[a,b]}var f;(f=google).x||(f.x=function(a,b){e(a,b,google.y)});var g;(g=google).sx||(g.sx=function(a,b){e(a,b,google.sy)});google.lm=[];var h;(h=google).plm||(h.plm=function(a){google.lm.push.apply(google.lm,a)});google.lq=[];var k;(k=google).load||(k.load=function(a,b,c){google.lq.push([a,b,c]);var l;(l=google).loadAll||(l.loadAll=function(a,b){google.lq.push([a,b]);google.bx=!1;var m;(m=google).lx||(m.lx=function()){var n=[];p;(p=google).fce||(p.fce=function(a,b,c,d){n.push([a,b,c,d]);google.qce=n;google.adl=[];}).call(this);google.f={};(function(){document.documentElement.addEventListener("submit",function(b){var a;if(a=b.target){var c=a.getAttribute("data-submit-false");a=c==="1"||c==="q"&&!a.elements.q.value?0:1}else a=!1;a&&(b.preventDefault(),b.stopPropagation()),!0);document.documentElement.addEventListener("click",function(b){var a;a={for(a=b.target;a&&a!==document.documentElement;a=a.parentElement)if(a.tagName==="A"){a=a.getAttribute("data-nohref")==="1";break a}a=!1;a&&b.preventDefault(),!0}}).call(this);</script><style>#gbar,#guser{font-size:13px;padding-top:1px !important;}#gbar{height:22px;}#guser{padding-bottom:7px !important;text-align:right}.gbh,.gbd{border-top:1px solid #c9d7f1;font-size:1px}.gbh{height:0;position:absolute;top:24px;width:100%}@media all{.gb1{height:22px;margin-right:.5em;vertical-align:top}#gbar{float:left}.gb1,a.gb4{text-decoration:underline !important}.gb1,a.gb4{color:#00c !important}.gb1 .gb4{color:#dd8e27 !important}.gbf .gb4{color:#900 !important}}</style><style>body,td,a,p,.h{font-family:sans-serif}body{margin:0;overflow-y:scroll}#gog{padding:3px 8px 0}td{line-height:.8em}.gac_m td{line-height:17px}form{margin-bottom:20px}.h{color:#1967d2}em{font-weight:bold;font-style:normal}.lst{height:25px;width:496px}.gsfi,.lst{font:18px sans-serif}.gsfs{font:17px sans-serif}.ds{display:inline-block;display:inline-block;margin:3px 0 4px;margin-left:4px}input{font-family:inherit}body{background:#fff;color:#1f1f1f}a{color:#681da8;text-decoration:none}a:active{color:#1967d2}a:visited{color:#681da8}.sblc{padding-top:5px}.sblc a{display:block;margin:2px 0;margin-left:13px;font-size:11px}.lsbb{background:#ecedee;border:solid 1px;border-color:#d2d2d2 #70757a #70757a #d2d2d2;height:30px}.lsbb{display:block}#WQoANb a{display:inline-block;margin:0 12px}.lsb{background:url(/images/nav_logo229.png) 0 -261px repeat-x;color:#1f1f1f;border:none;cursor:pointer;height:30px;margin:0;outline:0;font:15px sans-serif;vertical-align:top}.lsb:active{background:#dadce0}.lst:focus
```

Verificar Security Group desde el Servidor

Ver puertos en escucha (después de instalar servicios web)

```
sudo netstat -tulnp | grep LISTEN
```

O con ss (más moderno)

```
sudo ss -tulnp | grep LISTEN
```

```

))}}());</script></body></html>ubuntu@ip-172-31-74-199:~$ sudo netstat -tulpn | grep LISTEN
tcp        0      0 127.0.0.53:53          0.0.0.0:*        LISTEN      323/systemd-resolve
tcp        0      0 0.0.0.0:22             0.0.0.0:*        LISTEN      1/init
tcp        0      0 127.0.0.54:53          0.0.0.0:*        LISTEN      323/systemd-resolve
tcp6       0      0 :::22                  :::*             LISTEN      1/init
ubuntu@ip-172-31-74-199:~$

```

6. Solución de Problemas

Error: "Permission denied (publickey)"

Causas posibles:

1. Permisos incorrectos del archivo PEM
2. Usuario incorrecto
3. Clave PEM incorrecta

Soluciones:

1. Verificar y corregir permisos

```
ls -la ~/.ssh/labsuser.pem
```

```
chmod 400 ~/.ssh/labsuser.pem
```

2. Probar con usuario correcto

```
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

3. Usar modo verboso para diagnóstico

```
ssh -v -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

Error: "Connection refused"

Causas posibles:

1. Instancia no está en estado "Running"
2. Security Group no permite puerto 22
3. IP incorrecta

Soluciones:

Verificar que la instancia responde

```
ping TU-IP-PUBLICA
```

Verificar conectividad al puerto SSH

```
telnet TU-IP-PUBLICA 22
```

O con nc

```
nc -zv TU-IP-PUBLICA 22
```

Revisar Security Group en AWS Console

EC2 → Instances → Security tab → Security groups

Error: "Connection timed out"

Causas posibles:

1. Firewall local bloqueando conexión
2. Red corporativa/universitaria bloqueando SSH
3. Security Group mal configurado

Soluciones:

Probar con modo verboso

```
ssh -vvv -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

Si estás en red corporativa, consulta con tu administrador de red

Algunos entornos bloquean el puerto 22 saliente

Error: "Host key verification failed"

Causa: La instancia fue recreada con la misma IP pero diferente huella digital.

Solución:

Eliminar la entrada antigua

```
ssh-keygen -R TU-IP-PUBLICA
```

Volver a conectar

```
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

Usuario Incorrecto

Para Ubuntu: usa **ubuntu** Para Amazon Linux: usa **ec2-user** Para Debian: usa **admin**

Ubuntu (nuestro caso)

```
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

Amazon Linux

```
ssh -i ~/.ssh/labsuser.pem ec2-user@TU-IP-PUBLICA
```

Comandos de Referencia Rápida

Gestión de Claves PEM

Crear directorio SSH

```
mkdir -p ~/.ssh && chmod 700 ~/.ssh
```

```
# Copiar clave desde Windows
cp /mnt/c/Users/TU-USUARIO/Downloads/labsuser.pem ~/.ssh/
```

```
# Establecer permisos
chmod 400 ~/.ssh/labsuser.pem
```

```
# Verificar permisos
ls -la ~/.ssh/labsuser.pem
```

Conexión SSH

```
# Conexión básica
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

```
# Conexión con modo verboso (debug)
ssh -v -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA
```

```
# Ejecutar comando remoto sin sesión interactiva
ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA "comando"
```

```
# Desconectar
exit
# o
logout
# o presionar Ctrl+D
```

Verificación de Conectividad

```
# Ping a la instancia
ping -c 4 TU-IP-PUBLICA
```

```
# Verificar puerto SSH abierto
nc -zv TU-IP-PUBLICA 22
```

```
# Test de conectividad HTTPS
curl -I http://TU-IP-PUBLICA:8080
```

Próximos Pasos

Una vez conectado exitosamente:

Actualizar el sistema:

```
sudo apt update && sudo apt upgrade -y
```

1.

2. Instalar servidores web (Segunda Práctica):

- Apache en puerto 8080
- Nginx en puerto 8081
- Caddy en puerto 8082
- Apache HTTPS en puerto 8443

Verificar servicios desde el navegador:

`http://TU-IP-PUBLICA:8080`

`http://TU-IP-PUBLICA:8081`

`http://TU-IP-PUBLICA:8082`

`https://TU-IP-PUBLICA:8443`

Buenas Prácticas de Seguridad

Para la Clave PEM

- Guarda copias de seguridad cifradas
- Nunca la subas a GitHub o repositorios públicos
- Usa permisos 400 o 600
- Almacénala en `~/ .ssh/`
- Nunca la compartas por email o mensajería
- No uses la misma clave para múltiples proyectos críticos

Para Security Groups

- Usa grupos específicos por función
- Documenta cada regla
- Revisa reglas periódicamente
- No uses 0.0.0.0/0 en producción si puedes evitarlo
- No abras más puertos de los necesarios

Para las Instancias

- Mantén el sistema actualizado
- Instala solo software necesario
- Configura logs y monitoreo
- Haz backups regulares
- No ejecutes servicios como root
- No desactives el firewall

Recursos Adicionales

Documentación Oficial

- [AWS EC2 Documentation](#)
- [AWS Security Groups](#)
- [SSH Best Practices](#)

Comandos Útiles de Linux

- `man ssh` - Manual de SSH
- `man chmod` - Manual de permisos
- `man ls` - Manual del comando ls

Glosario

AMI (Amazon Machine Image): Plantilla que contiene el sistema operativo y configuración inicial de una instancia EC2.

EC2 (Elastic Compute Cloud): Servicio de AWS que proporciona capacidad de cómputo escalable en la nube.

PEM (Privacy Enhanced Mail): Formato de archivo que almacena claves criptográficas y certificados.

Security Group: Firewall virtual que controla el tráfico de red hacia y desde instancias EC2.

SSH (Secure Shell): Protocolo de red para acceso remoto seguro a sistemas.

VPC (Virtual Private Cloud): Red virtual privada aislada en AWS.

WSL (Windows Subsystem for Linux): Capa de compatibilidad para ejecutar Linux en Windows.

Repetición de práctica anterior (CONFIGURACIÓN AVANZADA DE SERVIDORES WEB Y HTTPS)

```
ubuntu@ip-172-31-74-199:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 2 not upgraded.
Need to get 2086 kB of archives.
After this operation, 8090 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [10
8 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7
[11.2 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B
]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.8 [13
31 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.8 [163
kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.8 [
97.7 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.8 [90.2 k
B]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2086 kB in 0s (34.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 103436 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../4-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../5-apache2-bin_2.4.58-1ubuntu8.8_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu8.8) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../6-apache2-data_2.4.58-1ubuntu8.8_all.deb ...
Unpacking apache2-data (2.4.58-1ubuntu8.8) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../7-apache2-utils_2.4.58-1ubuntu8.8_amd64.deb ...
```

```
ubuntu@ip-172-31-74-199:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 10:03:56 UTC; 1min 9s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 11548 (apache2)
      Tasks: 55 (limit: 1008)
     Memory: 5.3M (peak: 5.4M)
        CPU: 36ms
    CGroup: /system.slice/apache2.service
            └─11548 /usr/sbin/apache2 -k start
              └─11551 /usr/sbin/apache2 -k start
                └─11552 /usr/sbin/apache2 -k start

Dec 09 10:03:56 ip-172-31-74-199 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 09 10:03:56 ip-172-31-74-199 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-74-199:~$
```



```
ubuntu@ip-172-31-74-199:~$ sudo nano /etc/apache2/ports.conf
```

```
ubuntu@ip-172-31-74-199: ~
GNU nano 7.2 /etc/apache2/ports.conf *
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8080_

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```

```
ubuntu@ip-172-31-74-199:~$ sudo nano /etc/apache2/sites-available/000-default.conf
```

```
ubuntu@ip-172-31-74-199: ~
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:8080_
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```



```

ubuntu@ip-172-31-74-199:~$ sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache
  php8.3-readline
0 upgraded, 9 newly installed, 0 to remove and 2 not upgraded.
Need to get 4922 kB of archives.
After this operation, 22.4 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.5 [740 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.5 [371 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.5 [13.5 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cli amd64 8.3.6-0ubuntu0.24.04.5 [1915 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapache2-mod-php8.3 amd64 8.3.6-0ubuntu0.24.04.5 [1851 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libapache2-mod-php all 2:8.3+93ubuntu2 [4224 B]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3 all 8.3.6-0ubuntu0.24.04.5 [9174 B]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php all 2:8.3+93ubuntu2 [4076 B]
Fetched 4922 kB in 0s (67.7 MB/s)

```

```

ubuntu@ip-172-31-74-199:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 10:08:37 UTC; 41s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 18197 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 18201 (apache2)
    Tasks: 6 (limit: 1008)
   Memory: 10.6M (peak: 11.0M)
      CPU: 48ms
   CGroup: /system.slice/apache2.service
           └─18201 /usr/sbin/apache2 -k start
             └─18204 /usr/sbin/apache2 -k start
               └─18205 /usr/sbin/apache2 -k start
                 └─18206 /usr/sbin/apache2 -k start
                   └─18207 /usr/sbin/apache2 -k start
                     └─18208 /usr/sbin/apache2 -k start

Dec 09 10:08:37 ip-172-31-74-199 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 09 10:08:37 ip-172-31-74-199 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-74-199:~$

```

```

ubuntu@ip-172-31-74-199:~$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
<?php phpinfo(); ?>
ubuntu@ip-172-31-74-199:~$

```

```
ubuntu@ip-172-31-74-199:~$ curl http://localhost:8080/info.php
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"><head>
<style type="text/css">
body {background-color: #fff; color: #222; font-family: sans-serif;}
pre {margin: 0; font-family: monospace;}
a:link {color: #009; text-decoration: none; background-color: #fff;}
a:hover {text-decoration: underline;}
table {border-collapse: collapse; border: 0; width: 934px; box-shadow: 1px 2px 3px rgba(0, 0, 0, 0.2);}
.center {text-align: center;}
.center table {margin: 1em auto; text-align: left;}
.center th {text-align: center !important;}
td, th {border: 1px solid #666; font-size: 75%; vertical-align: baseline; padding: 4px 5px;}
th {position: sticky; top: 0; background: inherit;}
h1 {font-size: 150%;}
h2 {font-size: 125%;}
h2 a:link, h2 a:visited {color: inherit; background: inherit;}
.p {text-align: left;}
.e {background-color: #ccf; width: 300px; font-weight: bold;}
.h {background-color: #99c; font-weight: bold;}
.v {background-color: #ddd; max-width: 300px; overflow-x: auto; word-wrap: break-word;}
.v i {color: #999;}
img {float: right; border: 0;}
hr {width: 934px; background-color: #ccc; border: 0; height: 1px;}
:root {--php-dark-grey: #333; --php-dark-blue: #4F5B93; --php-medium-blue: #8892BF; --php-light-blue: #E2E4EF; --php-
accent-purple: #793862}@media (prefers-color-scheme: dark) {
  body {background: var(--php-dark-grey); color: var(--php-light-blue)}
  .h td, td.e, th {border-color: #606A90}
  td {border-color: #505153}
  .e {background-color: #404A77}
  .h {background-color: var(--php-dark-blue)}
  .v {background-color: var(--php-dark-grey)}
  hr {background-color: #505153}
}
</style>
<title>PHP 8.3.6 - phpinfo()</title><meta name="ROBOTS" content="NOINDEX,NOFOLLOW,NOARCHIVE" /></head>
<body><div class="center">
<table>
```

```
ubuntu@ip-172-31-74-199: ~  
</table>  
</div></body></html>ubuntu@ip-172-31-74-199:~$ sudo apt install certbot python3-certbot-apache -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  Augeas-lenses libaugeas0 python3-acme python3-augeas python3-certbot python3-configargparse python3-icu  
  python3-josepy python3-parsedatetime python3-rfc3339  
Suggested packages:  
  Augeas-doc python-certbot-doc python-certbot-nginx Augeas-tools python-acme-doc python-certbot-apache-doc  
The following NEW packages will be installed:  
  Augeas-lenses certbot libaugeas0 python3-acme python3-augeas python3-certbot python3-certbot-apache  
  python3-configargparse python3-icu python3-josepy python3-parsedatetime python3-rfc3339  
0 upgraded, 12 newly installed, 0 to remove and 2 not upgraded.  
Need to get 1657 kB of archives.  
After this operation, 8599 kB of additional disk space will be used.  
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Augeas-lenses all 1.14.1-1build2 [323 kB]  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 libaugeas0 amd64 1.14.1-1build2 [166 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-josepy all 1.14.0-1 [22.1 kB]  
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-rfc3339 all 1.1-4 [6744 B]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-acme all 2.9.0-1 [48.5 kB]  
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-augeas all 0.5.0-1.1 [9124 B]  
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-configargparse all 1.7-1 [31.7 kB]  
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-parsedatetime all 2.6-3 [32.8 kB]  
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-certbot all 2.9.0-1 [267 kB]  
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 certbot all 2.9.0-1 [89.2 kB]  
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-certbot-apache all 2.9.0-1 [128 kB]  
]  
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 python3-icu amd64 2.12-1build2 [534 kB]  
Fetched 1657 kB in 0s (32.8 MB/s)  
Preconfiguring packages ...  
Selecting previously unselected package Augeas-lenses.  
(Reading database ... 104291 files and directories currently installed.)  
Preparing to unpack .../00-augeas-lenses_1.14.1-1build2_all.deb ...  
Unpacking Augeas-lenses (1.14.1-1build2) ...  
Selecting previously unselected package libaugeas0:amd64.  
Preparing to unpack .../01-libaugeas0_1.14.1-1build2_amd64.deb ...  
Unpacking libaugeas0:amd64 (1.14.1-1build2) ...  
Selecting previously unselected package python3-josepy.  
Preparing to unpack .../02-python3-josepy_1.14.0-1_all.deb ...  
Unpacking python3-josepy (1.14.0-1) ...  
Selecting previously unselected package python3-rfc3339.  
Preparing to unpack .../03-python3-rfc3339_1.1-4_all.deb ...  
Unpacking python3-rfc3339 (1.1-4) ...  
Selecting previously unselected package python3-acme.  
Preparing to unpack .../04-python3-acme_2.9.0-1_all.deb ...  
Unpacking python3-acme (2.9.0-1) ...  
Selecting previously unselected package python3-augeas.  
Preparing to unpack .../05-python3-augeas_0.5.0-1.1_all.deb ...  
Unpacking python3-augeas (0.5.0-1.1) ...  
Selecting previously unselected package python3-configargparse.  
Preparing to unpack .../06-python3-configargparse_1.7-1_all.deb ...  
Unpacking python3-configargparse (1.7-1) ...  
Selecting previously unselected package python3-parsedatetime.  
Preparing to unpack .../07-python3-parsedatetime_2.6-3_all.deb ...  
Unpacking python3-parsedatetime (2.6-3) ...  
Selecting previously unselected package python3-certbot.  
Preparing to unpack .../08-python3-certbot_2.9.0-1_all.deb ...  
Unpacking python3-certbot (2.9.0-1) ...  
Selecting previously unselected package certbot.  
Preparing to unpack .../09-certbot_2.9.0-1_all.deb ...  
Unpacking certbot (2.9.0-1) ...
```



```
ubuntu@ip-172-31-74-199: ~
GNU nano 7.2 /etc/apache2/sites-available/default-ssl.conf *
<VirtualHost *:8443>
    ServerAdmin webmaster@localhost

    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

    #
    # SSL Engine Switch:
    # Enable/Disable SSL for this virtual host.
    SSLEngine on

    #
    # A self-signed (snakeoil) certificate can be created by installing
    # the ssl-cert package. See
    # /usr/share/doc/apache2/README.Debian.gz for more info.
    # If both key and certificate are stored in the same file, only the
    # SSLCertificateFile directive is needed.
    SSLCertificateFile /etc/ssl/certs/ssl-cert-snakeoil.pem
    SSLCertificateKeyFile /etc/ssl/private/ssl-cert-snakeoil.key

    #
    # Server Certificate Chain:
    # Point SSLCertificateChainFile at a file containing the
    # concatenation of PEM encoded CA certificates which form the
    # certificate chain for the server certificate. Alternatively
    # the referenced file can be the same as SSLCertificateFile
    # when the CA certificates are directly appended to the server
    # certificate for convinience.
    #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

    #
    # Certificate Authority (CA):
    # Set the CA certificate verification path where to find CA
    # certificates for client authentication or alternatively one
    # huge file containing all of them (file must be PEM encoded)
    # Note: Inside SSLCACertificatePath you need hash symlinks
    # to point to the certificate files. Use the provided
    # Makefile to update the hash symlinks after changes.
    #SSLCACertificatePath /etc/ssl/certs/
    #SSLCACertificateFile /etc/apache2/ssl.crt/ca-bundle.crt

    #
    # Certificate Revocation Lists (CRL):
    # Set the CA revocation path where to find CA CRLs for client
    # authentication or alternatively one huge file containing all
    # of them (file must be PEM encoded)
    # Note: Inside SSLCARevocationPath you need hash symlinks
    # to point to the certificate files. Use the provided

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^_ Replace    ^P Paste      ^J Justify    ^_ Go To Line M-E Redo
```

```
ubuntu@ip-172-31-74-199:~$ sudo nano /etc/apache2/ports.conf
```

```
ubuntu@ip-172-31-74-199: ~  
GNU nano 7.2 /etc/apache2/ports.conf *  
# If you just change the port or add more ports here, you will likely also  
# have to change the VirtualHost statement in  
# /etc/apache2/sites-enabled/000-default.conf  
  
Listen 8080  
Listen 8443_  
  
<IfModule ssl_module>  
    Listen 443  
</IfModule>  
  
<IfModule mod_gnutls.c>  
    Listen 443  
</IfModule>  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line   M-E Redo
```

```
ubuntu@ip-172-31-74-199:~$ sudo apt install nginx -y
```

```
ubuntu@ip-172-31-74-199:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Tue 2025-12-09 11:10:28 UTC; 1min 2s ago  
     Docs: man:nginx(8)  
   Process: 19155 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
   Process: 19156 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
  Main PID: 19193 (nginx)  
    Tasks: 3 (limit: 1008)  
   Memory: 2.4M (peak: 5.3M)  
      CPU: 27ms  
   CGroup: /system.slice/nginx.service  
           └─19193 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"  
             └─19196 "nginx: worker process"  
               └─19197 "nginx: worker process"  
  
Dec 09 11:10:28 ip-172-31-74-199 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:  
Dec 09 11:10:28 ip-172-31-74-199 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:  
lines 1-17/17 (END)
```

```
ubuntu@ip-172-31-74-199:~$ sudo nano /etc/nginx/sites-available/default
```

```
ubuntu@ip-172-31-74-199: ~  X  +  v  -  □  X
GNU nano 7.2 /etc/nginx/sites-available/default *
##
# You should look at the following URL's in order to grasp a solid understanding
# of Nginx configuration files in order to fully unleash the power of Nginx.
# https://www.nginx.com/resources/wiki/start/
# https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/
# https://wiki.debian.org/Nginx/DirectoryStructure
#
# In most cases, administrators will remove this file from sites-enabled/ and
# leave it as reference inside of sites-available where it will continue to be
# updated by the nginx packaging team.
#
# This file will automatically load configuration files provided by other
# applications, such as Drupal or Wordpress. These applications will be made
# available underneath a path with that package name, such as /drupal8.
#
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
##

# Default server configuration
#
server {
    listen 8081 default_server;
    listen [::]:8081 default_server;

    # SSL configuration
    #
    # listen 443 ssl default_server;
    # listen [::]:443 ssl default_server;
    #
    # Note: You should disable gzip for SSL traffic.
    # See: https://bugs.debian.org/773332
    #
    # Read up on ssl_ciphers to ensure a secure configuration.
    # See: https://bugs.debian.org/765782
    #
    # Self signed certs generated by the ssl-cert package
    # Don't use them in a production server!
    #
    # include snippets/snakeoil.conf;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.htm index.nginx-debian.html;

    server_name _;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/ Go To Line  M-E Redo

ubuntu@ip-172-31-74-199:~$ echo "<h1>Servidor Nginx</h1><p>Funcionando en puerto 8081</p>" | sudo tee
/usr/share/nginx/html/index.html
<h1>Servidor Nginx</h1><p>Funcionando en puerto 8081</p>
```



```

ubuntu@ip-172-31-74-199:~$ sudo systemctl restart nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload units.
ubuntu@ip-172-31-74-199:~$ sudo systemctl status nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run '
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-10 11:53:39 UTC; 21s ago
     Docs: man:nginx(8)
  Process: 1730 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, s
  Process: 1733 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/S
 Main PID: 1734 (nginx)
    Tasks: 3 (limit: 1017)
   Memory: 2.4M (peak: 2.6M)
      CPU: 20ms
   CGroup: /system.slice/nginx.service
           └─1734 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1735 "nginx: worker process"
               └─1736 "nginx: worker process"

Dec 10 11:53:39 ip-172-31-74-199 systemd[1]: Starting nginx.service - A high performance web server a
Dec 10 11:53:39 ip-172-31-74-199 systemd[1]: Started nginx.service - A high performance web server an
lines 1-18/18 (END)

```

```

ubuntu@ip-172-31-74-199:~$ sudo netstat -tulpn | grep nginx
tcp        0      0 0.0.0.0:8081        0.0.0.0:*          LISTEN     1734/nginx: master
tcp6       0      0 :::8081            :::*                LISTEN     1734/nginx: master
ubuntu@ip-172-31-74-199:~$ curl http://localhost:8081
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1
-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

      div.main_page {
        position: relative;
        display: table;

        width: 800px;

        margin-bottom: 3px;
        margin-left: auto;
        margin-right: auto;
        padding: 0px 0px 0px 0px;

        border-width: 2px;
        border-color: #212738;
        border-style: solid;

        background-color: #FFFFFF;

```



```

ubuntu@ip-172-31-74-199:~$ sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
The following NEW packages will be installed:
  apt-transport-https debian-archive-keyring debian-keyring
0 upgraded, 3 newly installed, 0 to remove and 2 not upgraded.
Need to get 31.5 MB of archives.
After this operation, 33.4 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https
all 2.8.3 [3970 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 debian-archive-keyring all 2
023.4ubuntu1 [168 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 debian-keyring all 2023.12.2
4 [31.3 MB]
Fetched 31.5 MB in 1s (34.0 MB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 105385 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Selecting previously unselected package debian-archive-keyring.
Preparing to unpack .../debian-archive-keyring_2023.4ubuntu1_all.deb ...
Unpacking debian-archive-keyring (2023.4ubuntu1) ...
Selecting previously unselected package debian-keyring.
Preparing to unpack .../debian-keyring_2023.12.24_all.deb ...
Unpacking debian-keyring (2023.12.24) ...
Setting up apt-transport-https (2.8.3) ...
Setting up debian-archive-keyring (2023.4ubuntu1) ...
Setting up debian-keyring (2023.12.24) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

```

ubuntu@ip-172-31-74-199:~$ curl -sLf 'https://dl.cloudsmith.io/public/caddy/stable/gpg.key' | sudo gp
g --dearmor -o /usr/share/keyrings/caddy-stable-archive-keyring.gpg
ubuntu@ip-172-31-74-199:~$ curl -sLf 'https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt' |
sudo tee /etc/apt/sources.list.d/caddy-stable.list
# Source: Caddy
# Site: https://github.com/caddyserver/caddy
# Repository: Caddy / stable
# Description: Fast, multi-platform web server with automatic HTTPS

deb [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/public/c
addy/stable/deb/debian any-version main

deb-src [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/publ
ic/caddy/stable/deb/debian any-version main
ubuntu@ip-172-31-74-199:~$

```

```

ubuntu@ip-172-31-74-199:~$ sudo apt update && sudo apt install caddy -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version InRelease [14.8 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1646 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [307 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.8 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1501 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [304 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.3 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2350 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [535 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7124 B]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:22 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version/main amd64 Packages [4329 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [915 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [206 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:28 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 8875 kB in 2s (4757 kB/s)

```

```

ubuntu@ip-172-31-74-199:~$ sudo mkdir -p /var/www/caddy
ubuntu@ip-172-31-74-199:~$ echo "# Bienvenido a Caddy" | sudo tee /var/www/caddy/README.md
# Bienvenido a Caddy
ubuntu@ip-172-31-74-199:~$ echo "" | sudo tee -a /var/www/caddy/README.md

ubuntu@ip-172-31-74-199:~$ echo "Este servidor está funcionando correctamente." | sudo tee -a /var/www/caddy/README.md
Este servidor está funcionando correctamente.
ubuntu@ip-172-31-74-199:~$ echo "" | sudo tee -a /var/www/caddy/README.md

ubuntu@ip-172-31-74-199:~$ echo "## Características" | sudo tee -a /var/www/caddy/README.md
## Características
ubuntu@ip-172-31-74-199:~$ echo "- Servidor moderno" | sudo tee -a /var/www/caddy/README.md
- Servidor moderno
ubuntu@ip-172-31-74-199:~$ echo "- HTTPS automático" | sudo tee -a /var/www/caddy/README.md
- HTTPS automático
ubuntu@ip-172-31-74-199:~$ echo "- Fácil configuración" | sudo tee -a /var/www/caddy/README.md
- Fácil configuración
ubuntu@ip-172-31-74-199:~$

```

```

ubuntu@ip-172-31-74-199:~$ curl -o /tmp/test-image.jpg "https://www.python.org/static/apple-touch-icon-144x144-precomposed.png"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 7382 100 7382    0     0  118k    0 --:--:-- --:--:-- --:--:-- 120k
ubuntu@ip-172-31-74-199:~$ sudo mv /tmp/test-image.jpg /var/www/caddy/test.jpg
ubuntu@ip-172-31-74-199:~$

```

```

ubuntu@ip-172-31-74-199:~$ sudo nano /etc/caddy/Caddyfile

```

ubuntu@ip-172-31-74-199: ~

GNU nano 7.2

/etc/caddy/Caddyfile *

```
# The Caddyfile is an easy way to configure your Caddy web server
#
# Unless the file starts with a global options block, the first
# uncommented line is always the address of your site.
#
# To use your own domain name (with automatic HTTPS), first make
# sure your domain's A/AAAA DNS records are properly pointed to
# this machine's public IP, then replace ":80" below with your
# domain name.

:8082 {
    # Set this path to your site's directory.
    root * /usr/www/caddy

    # Enable the static file server.
    file_server browse

    @markdown path *.md
    header @markdown Content-Type text/plain
    # Another common task is to set up a reverse proxy:
    # reverse_proxy localhost:8080

    # Or serve a PHP site through php-fpm:
    # php_fastcgi localhost:9000
}

# Refer to the Caddy docs for more information:
# https://caddyserver.com/docs/caddyfile
```

```

ubuntu@ip-172-31-74-199:~$ sudo systemctl restart caddy
ubuntu@ip-172-31-74-199:~$ sudo systemctl status caddy
● caddy.service - Caddy
   Loaded: loaded (/usr/lib/systemd/system/caddy.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-10 12:02:34 UTC; 5s ago
     Docs: https://caddyserver.com/docs/
   Main PID: 2667 (caddy)
    Tasks: 7 (limit: 1017)
   Memory: 10.7M (peak: 11.1M)
      CPU: 75ms
   CGroup: /system.slice/caddy.service
           └─2667 /usr/bin/caddy run --environ --config /etc/caddy/Caddyfile

Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.2746327,"logger":"admin">
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.2749531,"logger":"tls.c>
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"warn","ts":1765368154.2749588,"logger":"http">
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"warn","ts":1765368154.2749774,"logger":"http">
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.274981,"logger":"http.l>
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.2751155,"msg":"autosave>
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.2751694,"msg":"servin>
Dec 10 12:02:34 ip-172-31-74-199 systemd[1]: Started caddy.service - Caddy.
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.279612,"logger":"tls", ">
Dec 10 12:02:34 ip-172-31-74-199 caddy[2667]: {"level":"info","ts":1765368154.2797005,"logger":"tls", ">
ubuntu@ip-172-31-74-199:~$ sudo netstat -tulpn | grep caddy
tcp        0      0 127.0.0.1:2019      0.0.0.0:*        LISTEN      2667/caddy
tcp6       0      0 :::8082             :::*             LISTEN      2667/caddy
ubuntu@ip-172-31-74-199:~$

```